



Cradle Care - Smart IoT Baby Monitoring System

Official Mobile Application: **Cradle Care**

Division: **M**

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1. Team Members:

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- 2) Ghagas Shubham Shekhar – 16014125017
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2. Final Problem Statement:

To develop an IoT-integrated automated baby cradle that identifies infant distress (crying) and hygiene issues (wetness) while providing parents with a remote dashboard to monitor activities, receive alerts, and manually control the rocking motion from any location.

3. Proposed idea:

The proposed Smart E-Cradle system uses an ESP32 microcontroller as a smart wireless control unit integrated with multiple sensors and a mobile application. The system is designed to improve infant monitoring by combining automation, IoT connectivity, and remote accessibility in a compact cradle solution.

The project features:

- **Smart Auto-Swing Functionality** that automatically activates the cradle movement when abnormal conditions are detected.
- **Remote Mobile App Control** allowing parents to operate and monitor the cradle wirelessly.
- **Real-Time Activity Tracking** with live event updates and timestamp-based monitoring.
- **IoT-Based Cloud Connectivity** using Firebase for seamless communication between hardware and application.
- **Modern Smart-Cradle Design** focused on safety, convenience, and reduced parental workload.

Workflow: ESP32 → Firebase → Cradle Care App → Notifications

4. Cradle Care Mobile Application:

The Cradle Care mobile application acts as the central monitoring and control interface of the Smart E-Cradle system. It enables parents to remotely monitor the baby's condition in real time and receive instant alerts whenever abnormal activity is detected.

The application is developed using **React Native (Expo)** and integrated with **Firestore** for cloud communication and real-time data synchronization.

The application backend is deployed on **Render cloud platform** to enable real-time remote communication and live data synchronization.

Key Features

- **Real-Time Monitoring**
Displays live baby activity updates received from the ESP32 module through Firestore.
- **Cry Detection Alerts**
Sends instant push notifications when the sound sensor detects continuous crying above the defined threshold.
- **Wet Diaper Detection**
Detects moisture changes using the capacitive moisture sensor and alerts parents immediately.
- **Automatic Cradle Swinging**
Automatically activates the servo motor to gently swing the cradle when crying is detected.
- **Manual Cradle Control**
Allows users to manually start or stop cradle swinging directly from the mobile app dashboard.
- **Activity Dashboard**
Shows recent system events with timestamps for easier monitoring and tracking.
- **Cloud Connectivity**
Uses Firestore Realtime Database and Firestore Cloud Messaging (FCM) for communication between hardware and application.
- **User-Friendly Interface**
Designed with a clean and minimal UI for quick access to baby status and controls.

Application Workflow

1. Sensors connected to the ESP32 continuously monitor sound and moisture levels.
2. Sensor data is processed by the ESP32 microcontroller.
3. The ESP32 uploads activity data to Firestore in real time.
4. The Cradle Care app fetches and displays updated information instantly.
5. Firestore Cloud Messaging sends emergency notifications to the parent device when abnormal conditions are detected.
6. Parents can remotely control cradle movement using the app interface.

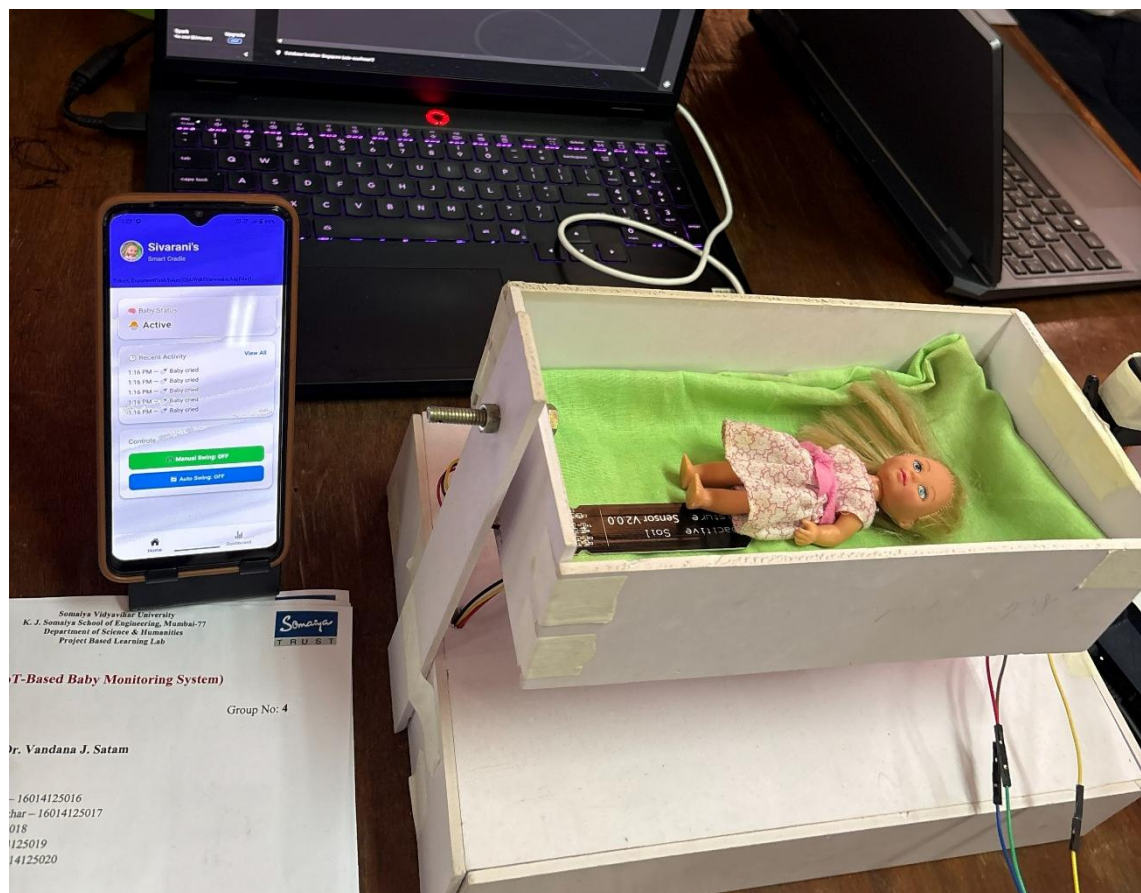
Technologies Used

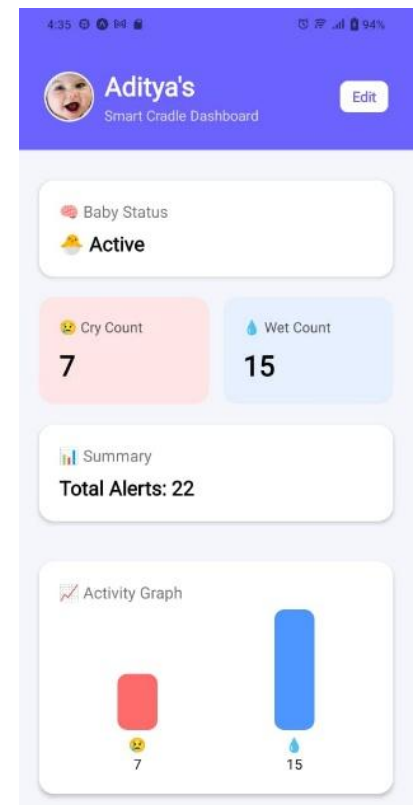
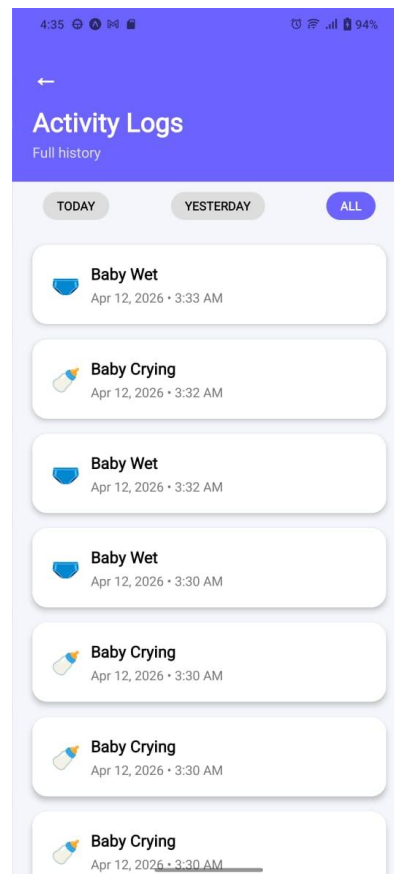
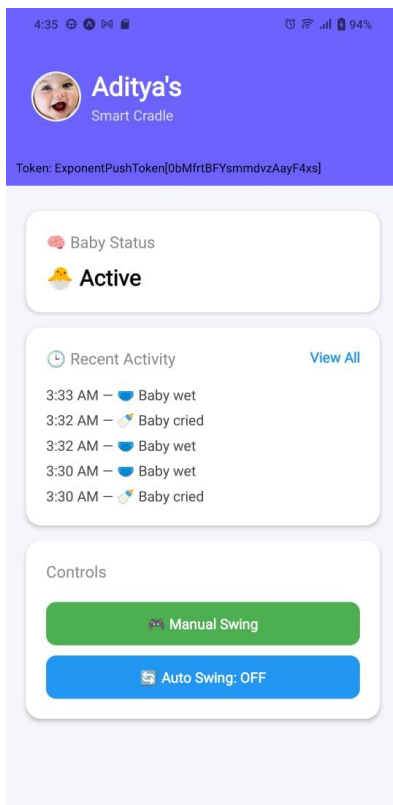
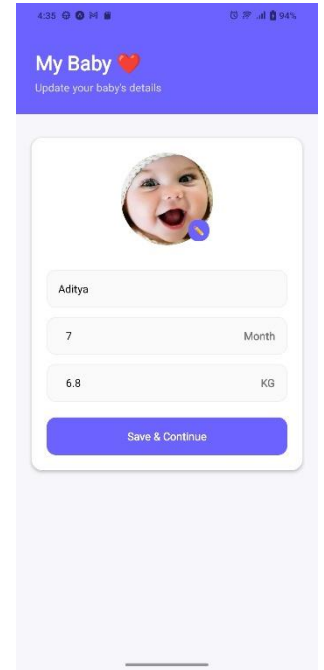
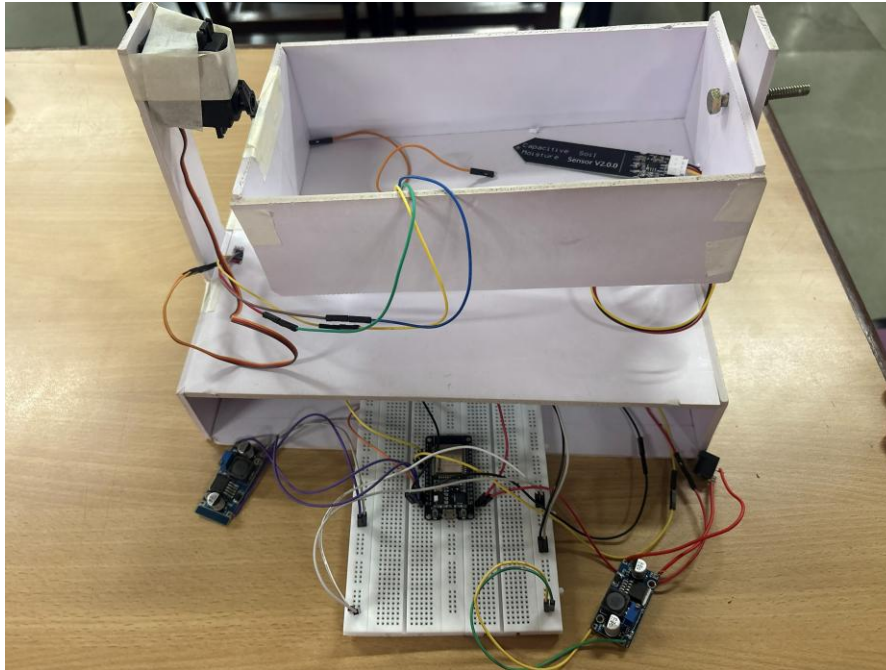
- React Native (Expo)
- Firebase Realtime Database
- Firebase Cloud Messaging (FCM)
- ESP32 Microcontroller
- Arduino IDE
- JavaScript

Future Enhancements

- AI-based cry pattern analysis
- Live baby camera streaming
- Sleep monitoring analytics
- Voice assistant integration
- Health report generation
- Multi-user parent access system

5. Photo of Developed Solution (Prototype & Mobile Application):





6. Total Expenditure:

Components	Quantity	Cost in ₹
ESP32 Microcontroller	1	400
MG996R Servo Motor	1	300
MAX9814 Sound Sensor Module	1	250
Capacitive Moisture Sensor Module	1	50
LM2596 Buck Convertor	2	160
DC barrel jack	1	5
Breadboard	2	120
Jumper Wires	1	160
12V 2A DC-AC Power Adapter	1	180
Multimeter	1	180
PVC Foam Sheet	1	640
Total		₹ 2445.00

7. Team Contribution:

Name	Contribution
Aditya S. Gadge	Lead Mobile App Development & Firebase Integration, Blender 3D modeling, documentation
Shubham S. Ghagas	Arduino/ESP32 coding, Virtual simulation, hardware assembly
Yug Goyal	Hardware procurement, Prototype simulation, hardware assembly
Saksham Gupta	Raw material procurement, Prototype Design & Development, prototype finalization
Shivrani Iggagunta	Prototype concept ideation, Prototype Design & Development, prototype finalization

8. Project Resources:

<https://drive.google.com/drive/folders/1gYv1jYx4v3EfwsTmthLoUL1IOzylz95y?usp=sharing>

GitHub Repository:

<https://github.com/adityagadge-dev>